

Question	Marking guidance	Mark	Comments
08.1	$T\sin 40$ or $T\cos 50$ seen in numbers or symbols ✓ $1.5(4) \times 10^8$ (N) ✓	2	Condone lack of factor of 2 for 1 st mark
08.2	Converts mass of tower into weight ✓ $2.2(4) \times 10^8$ (N) OR Adds correctly <u>weight</u> of tower to their 08.1 ✓	2	eg 7.1×10^6 g seen
08.3	$E = \frac{\sigma}{\epsilon}$ used ✓ $\sigma = \frac{F}{A}$ and $\epsilon = \frac{e}{l}$ used ✓ 0.41(3) (m)	3	Use of $E = \frac{Fl}{eA}$ gets both of the first two marks
08.4	Breaking stress – force per unit area (when the cable breaks) ✓ Plastic deformation – idea that change in shape is permanent OR extension is unrecoverable ✓	2	Not simply inelastic
08.5	Allows margin for error OR yield occurs at lower stresses OR can cope with unexpectedly large loads	1	Accept to prevent going into the plastic deformation region (of the stress-strain graph)
Total		10	

Question	Marking guidance	Mark	Comments
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04.1	$\frac{\text{extension}}{\text{unstretched length}}$	1	Accept $\frac{\text{extension}}{\text{original length}}$ OR $\frac{l}{\Delta l}$ with symbols defined OR extension per unit original length
04.2	Correctly indicated limit of proportionality (where the line stops being straight), either on the curve or on the y-axis <u>labelled L</u> ✓	1	
04.3	First property correct ✓ Second property correct ✓ Correct explanation of one property ✓	Max 3	Accept references to breaking stress, Young modulus, stiffness, elastic limit, limit of proportionality, yield Explanations must be in appropriate technical language; do not accept “easier to break”
Total		5	

Question	Marking guidance	Mark	Comments

Question	Marking guidance	Mark	Comments
10.1	0.8 and 1.9 ✓cao	1	1 dp for both answers
10.2	Both points accurately plotted ✓ Well-drawn line of best fit ✓	2	Points must be plotted within ½ small square of true position. The line of best fit should follow the trend of the points with an even scatter of points on either side of the line. Do not allow thick line or plots.
10.3	Points taken from large gradient triangle ✓ k calculated to be in the range $(1.08-1.17) \times 10^3$ ✓	2	Triangle hypotenuse must be greater than ½ length of candidate line. Gradient readings must be notated on the graph in some way for MP1. Accept 2 or 3 sf only
10.4	Use of $E = kL/A$ OR $E = \frac{FL}{A\Delta L}$ ✓ Young modulus = $(1.08 - 1.17) \times 10^{11}$ (Pa) ✓	2	ecf and no sf penalty
Total		7	

Question	Marking guidance	Mark	Comments
	reasonable precaution ✓ comments on precaution in terms of irradiation or contamination ✓	2	<i>Eg:</i> Use long tongs ✓ To reduce irradiation to the body ✓ OR

13	B
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14	C
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23	C
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Question	Marking guidance	Mark	Comments
01.1	Beta- <u>plus</u> or positron (emission) ✓	1	Accept ${}^0_{+1}\beta$ Accept correct decay equation in which beta plus is seen Just beta decay is insufficient
01.2	(electron) neutrino ✓	1	Accept correct symbols Reject anti-neutrino.
01.3	the correct antiparticle to student's 01.2. ✓	1	Accept correct symbols Expect (electron) antineutrino but not if 01.2 is incorrect ecf from 01.2

Question	Marking guidance	Mark	Comments
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Question	Marking guidance	Mark	Comments
11.1	Both points accurately plotted ✓ Well drawn straight line of best fit ✓	2	The line should accurately follow the trend of the points with an even scatter of points about the line. An intercept outside the range of 2.2 to 2.5 is not likely to be correct
11.2	It (the straight line) should go through the origin or (0,0) ✓	1	Not just “doesn’t go through zero”
11.3	The weight of the tray (has been ignored) ✓	1	Accept zero error (on ruler) but not inaccurate ruler or ruler not vertical – explanation has to be valid for a systematic error not for a random error
11.4	Gradient in the range 18.5 to 20.5 from a large triangle (Δl at least 7 mm) ecf for BFL ✓ Unit = N mm^{-1} or N m^{-1} consistent with the candidate's value ✓	2	Triangle need not be seen on this occasion
11.5	(The graph can be used to find a value for k) k is the gradient of the graph ✓ Since the systematic error does not affect the gradient of the graph or systematic error only affects the intercept. ✓	2	Accept correct reference to $y = mx + c$

Question	Marking guidance	Mark	Comments
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22	D
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23	B
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	Systematic: Incorrectly calibrated ruler OR zero error (ruler with zero mark not at the end or worn end) ✓	Mark each answer separately	
02	Max 3 from the following: • Photon interacts with another particle / nucleus ✓ • Photon transfers energy ✓ • Positron (and electron) formed ✓ • (Original) photon was gamma / High energy ✓ • Positron and electron move with equal speeds in opposite directions ✓	Accept neutral boson for photon Accept high-energy photon for gamma photon Can be seen as an equation; condone missing KE	3

Question	Marking guidance	Mark	Comments
07.1	$v = (20 \sin 15^\circ) = 5.2 \text{ (5.18) m s}^{-1} \checkmark$	1	

Question	Marking guidance	Mark	Comments
07.2	Correct use of equation of uniform motion $v^2 = u^2 + 2as \checkmark$ Correct substitution to give a value of 1.37 m $\checkmark$ Adds 1 m to give 2.37 m $\checkmark$	3	ecf incorrect value from 07.1 for velocity For MP1 more than half of substitutions must be correct. Condone rounding errors in MP2 Accept 2.38 m

Question	Marking guidance	Mark	Comments
07.3	Correct calculation of time to centre of the motion 0.53 s OR Calculates time to fall from centre to the ground 0.70 (0.695) s $\checkmark$ Total time of flight (= 0.52 + 0.69) = 1.22 s $\checkmark$ Horizontal distance = ($v_h t = 20 \cos 15^\circ \times 1.22$) = 23.5 m $\checkmark$	3	If no other marks awarded, allow 1 mark for adding together their two different times of flight.

Question	Marking guidance	Mark	Comments
07.4	$\frac{1}{2} 9.3 \times 10^3 \times 0.05^2 = 11.625 \text{ J} \checkmark$ use of $\frac{1}{2} k \times 0.05^2 = mgh \checkmark$ $h = 20.4(3) \text{ m} \checkmark$	3	If no other mark given, allow 1 mark for calculating the initial force from spring (465 N) Alternative MP2 (KE approach): Use of suvat and KE equation

Question	Marking guidance	Mark	Comments
07.5	In the spring: Not all of the energy stored in the spring is transferred to the ball wtte $\checkmark$ In the air: Work is done against air resistance (reducing the final PE of the ball) $\checkmark$	2	It must be clear whether answers refer to the spring or movement through the air. Alternative for MP2: Air resistance increases downwards acceleration of ball (reducing height reached).

Total		12
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Question	Marking guidance	Mark	Comments
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Question	Marking guidance	Mark	Comments
10.1	Well drawn line of best fit which, starting at the origin, is initially linear and then curves smoothly ✓	1	Overall, the line should accurately follow the trend of the points with an even scatter of points about the line Penalise multiple or thick lines

Question	Marking guidance	Mark	Comments
10.2	Clear attempt to measure the gradient at or below $1.4 \times 10^8 \text{ Pa}$ ✓ Answer in the range $(0.9\text{--}1.1) \times 10^{11}$ ✓	2	Do not penalise sf in MP1. There must be some evidence that a single point has not been used. Final answer 2 or 3 sf only

Question	Marking guidance	Mark	Comments
10.3	Establish the difference maximum and minimum gradients (consistent with the points) (along the linear part of the graph). ✓	1	OWTTE Allow discussion involving 'worst fit' line Allow difference between the worst of max OR min gradient and the best fit line gradient.

Question	Marking guidance	Mark	Comments
10.4	Strain calculated = 4×10^{-3} ✓ Estimate for number of 'cm' squares in the range 106–112 OR Clear attempt to convert squares to J m^{-3} seen ✓ Volume = $1.5 \times 0.397 \times 10^{-6}$ ✓ Answer in the range 0.30–0.35 (J) ✓	4	26–28 if '2 cm' squares counted. Allow ecf for square count if wrong strain value used. Unit required for final answer but no sf penalty Look for 5.1 to $5.8 \times 10^5 \text{ J m}^{-3}$ if worked out directly. Condone pot error in MP3 Candidates who use a strain value of 6×10^{-3} can be awarded 3 marks: Expect to see 184 to 191 'cm' squares or 45 to 48 '2 cm' squares to give 9.0 to $9.6 \times 10^5 \text{ J m}^{-3}$ And answer in range 0.53 to 0.58 J
Total		8	

Question	Key	Answer
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16	B	elastic	elastic
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Question	Answers	Additional comments/Guidelines	Mark	AO
06.1	Use of $F = k\Delta l$ ✓ Use of $F = mg$ ✓ 470 (N m ⁻¹) ✓	“Use of” means by manipulation or substitution. Accept answers that round to 470 Missing “x 13” is penalised in MP3	3	1 × AO1 2 × AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
06.2	Use of $E = \frac{1}{2}F\Delta l$ ✓ 6.5 (J) ✓ ecf for their k from 06.1	Accept use of $E = \frac{1}{2}k\Delta l^2$	2	AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
06.3	Idea that plates will compress spring more ✓ ... so that plates (will not be able to be removed as they) will be below the top of the ‘well’ ✓		2	AO3
Total			7	

Question	Answers	Additional comments/Guidelines	Mark	AO
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Question	Answers	Additional comments/Guidelines	Mark	AO
09.1	Uses $W = Fd$ ✓ 720×2.4 leading to 1728 (J) to at least 3 sf ✓	for MP2 must see 2.4 or 2 x 1.2 written.	2	1 × AO1 1 × AO3

Question	Answers	Additional comments/Guidelines	Mark	AO
09.2	Uses efficiency = $\frac{\text{useful work done}}{\text{total energy input}}$ or useful work done = $140 \times 9.81 \times 1.2$ ✓ 0.95 or 95 % ✓ ecf	97% or 0.97 if candidate uses 1700 (from 09.1)	2	1 × AO1 1 × AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
09.3	Calculate difference between work done on engine and work done by operator. ✓ Uses difference in work done = force x distance to determine weight of pulley. ✓	MP2 is contingent on correct method for MP1 Work done lifting pulley = candidate's 09.1 – 1650 $(\text{Weight} = \frac{80}{1.2}) = 67 \text{ (N)}$ ecf from 09.2 Allow for rounding Allow alternatives e.g. approach to say that $140 \text{ N} = 0.954 \times \text{total weight of engine and bottom pulley}$	2	1 × AO1 1 × AO3

Question	Answers	Additional comments/Guidelines	Mark	AO
09.4	Uses $\varepsilon = \frac{\Delta L}{L}$ and $\sigma = \frac{F}{A}$ and Young modulus = $\frac{\sigma}{\varepsilon}$ OR $e = \frac{FL}{EA}$ ✓ 2.3×10^{-2} (m) ✓	Award MP1 and MP2 for candidate who goes directly to $e = \frac{FL}{EA}$	2	1 × AO1 1 × AO2
Total			8	

Question	Answers	Additional comments/Guidelines	Mark	AO
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21	D	0.37 m	AO2
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Question	Answers	Additional comments/Guidelines	Mark	AO
07.1	use of $M = \rho V$ AND $V = AL$ ✓ 720 (kg m ⁻³) ✓	condone POT error for MP1	2	AO1 × 2

Question	Answers	Additional comments/Guidelines	Mark	AO
07.2	$E = \frac{1}{2}kx^2$ AND $E = mgh$ seen and equated, with 0.6 seen anywhere in the equation ✓ both combined correctly ✓	reject answers where 0.6 (or 60%) is not seen, e.g. 0.3 seen with no reasoning	2	AO2 × 2

Question	Answers	Additional comments/Guidelines	Mark	AO
07.3	use of $h \propto x^2$ OR use of $h = \frac{0.3 kx^2}{mg}$ (to determine k) ✓ 6.1 (cm) ✓	$k = 72 \text{ N cm}^{-1}$ correct final answer scores full marks	2	AO2 × 2

Question	Answers	Additional comments/Guidelines	Mark	AO
07.4	the height is double for R ✓ because the (elastic) energy stored is the same and $GPE = mgh$ (and mass has halved) ✓	for MP2 accept a correct equation relating mass to height and explicit comment that the other variables are constant	2	AO2 × 2
Total			8	

Question	Answers	Additional comments/Guidelines	Mark	AO
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Question	Answers	Additional comments/Guidelines	Mark	AO
10.1	$\Delta L = 9.6 \text{ mm}$ ✓	cao	1	AO1

Question	Answers	Additional comments/Guidelines	Mark	AO
10.2	correct plotting of their value ✓		1	AO3

Question	Answers	Additional comments/Guidelines	Mark	AO
10.3	best-fit line ✓ determination of gradient from points on line far apart (large triangle)✓ 410 to 425 (kg m^{-1}) ✓	2 or 3 sf	3	AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
10.4	uses <i>Young modulus</i> = $\frac{\text{gradient} \times g \times \text{length}}{\text{cross-sectional area}}$ or $F = mg$ ✓ value for Young modulus 1.0×10^{11} to 1.1×10^{11} Pa ✓	2 or 3 sf	2	AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
10.5	larger extension ✓ ₁ reduces (percentage) uncertainty in extension and therefore in E ✓ ₂ OR smaller cross-sectional area ✓ ₁ increases (percentage) uncertainty in c-s area and therefore in E ✓ ₂	accept greater strain for extension accept greater stress reject just 'area'	2	AO3
Total			9	

Question	Key	Answer	AO
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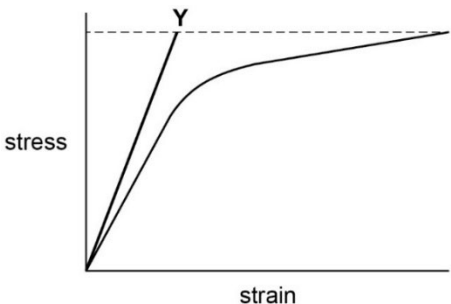
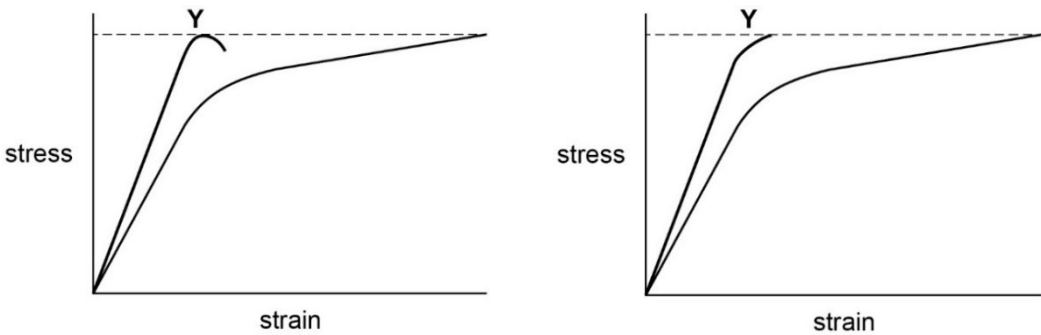
17	D	It undergoes little plastic deformation before breaking.	AO1
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18	B	the energy stored per unit volume	AO1
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19	A	$\frac{1}{12}$	AO1
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Question	Answers	Additional comments/Guidelines	Mark	AO
01	$\text{kg m}^{-1} \text{s}^{-2}$ ✓	units in any order Do not condone K for k	1	AO1
Total			1	

Question	Answers	Additional comments/Guidelines	Mark	AO
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Question	Answers	Additional comments/Guidelines	Mark	AO
03	straight line with steeper gradient and small or no reduction of gradient ✓ same max stress ✓	 stress strain for MP1: - Do not condone line that carries on towards P - Line must begin straight but may be curved near the top as shown but curved part must be less than half a grid square:  stress strain stress strain	2	AO1 × 2
Total			2	

Question	Answers	Additional comments/Guidelines	Mark	AO
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Question	Answers	Additional comments/Guidelines	Mark	AO
04.1	$(2)T\cos 30$ OR $(2)T\sin 60$ seen ✓ $15(.3) \text{ N}$ ✓	For MP1 allow omission of the factor of 2	2	AO1 × 2

Question	Answers	Additional comments/Guidelines	Mark	AO
04.2	use of $\sigma = \frac{F}{A}$ OR use of $E = \frac{\text{their } \sigma}{\epsilon}$ OR $\frac{\text{Their } \mathbf{04.1}}{0.75 \times 10^{-6} \times 1.5 \times 10^{-2}}$ ✓ $1.4 \times 10^9 \text{ (Pa)}$ ecf from 04.1 ✓	condone POT error in MP1 answer should be candidate's $\mathbf{04.1} \times 8.9 \times 10^7$	2	AO1 AO2
Total			4	

Question	Answers	Additional comments/Guidelines	Mark	AO
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Question	Answers	Additional comments/Guidelines	Mark	AO
07.1	use of kinematic equation(s) e.g. $s = ut + \frac{1}{2}at^2$ to find expression for t ✓ rearranges $E = \frac{1}{2}mv^2$ to get horizontal velocity ✓ shows both and multiplies them to get $D = 2\sqrt{\frac{HE}{mg}}$ ✓	If no other mark given, award 1 mark if $E = \frac{1}{2}mv^2$ AND $H = \frac{1}{2}gt^2$ AND $D = vt$ seen	3	AO1 × 3

Question	Answers	Additional comments/Guidelines	Mark	AO
07.2	no air resistance ✓	accept statement that the ball is not rotating condone “air friction” for “air resistance”	1	AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
07.3	obtains gradient = 14 ✓ correct rearrangement OR substitution into $D = \Delta L \sqrt{\frac{2Hk}{mg}}$ ✓ 36 (N m ⁻¹) ✓	Accept a single data point substituted into given equation for MP1 Allow POT error for MP1 and MP2 Accept answer that rounds to 36	3	AO3 × 3

Total			7	
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Question	Answers	Additional comments/Guidelines	Mark	AO
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Question	Answers	Additional comments/Guidelines	Mark	AO
09.1	Uses gradient of the graph to determine the Young modulus (188 GPa) including clear manipulation of powers of 10 ✓	Condone use of a single point (eg 0.0032, 600×10^6)	1	AO3

Question	Answers	Additional comments/Guidelines	Mark	AO
09.2	Rotate the turnbuckle so that the screws move closer together which will increase the extension of /strain in the cable ✓		1	AO4

Question	Answers	Additional comments/Guidelines	Mark	AO
09.3	Determines pitch $\left(\frac{1.3}{2}\right)$ and calculates ratio ✓	Expect to see 22 or 21.5	1	AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
09.4	Determines initial strain = 0.00105 to 0.00115 ✓	allow a calculation using the Young modulus and stress	1	AO3

Question	Answers	Additional comments/Guidelines	Mark	AO
09.5	Determines additional strain = 0.0058 ✓ strain = 0.0068 to 0.0069 ✓	expect to see $\frac{8 \times 1.3 \times 10^{-3}}{1.8}$	2	AO1 × 1 AO3 × 1

Question	Answers	Additional comments/Guidelines	Mark	AO
09.6	Use of stress = $\frac{F}{A}$ OR $E = \frac{\sigma}{\epsilon}$ ✓ Determines the stress as candidate's 09.5 × 1.9 × 10 ¹¹ ✓ T = 4.0 × 10 ⁴ to 5.1 × 10 ⁴ (N) ✓	No credit for answers based on 210 MPa Allow a value of stress read from graph consistent with their 09.5 Expect 1.28 × 10 ⁹ (Pa) allow ecf from 09.5	3	AO1 × 1 AO2 × 2
Total			9	

Question	Answers	Additional comments/Guidelines	Mark	AO
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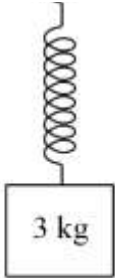
Question	Answers	Additional comments/Guidelines	Mark	AO
10.1	Calculates mean value of d OR half range ✓ Calculates mean value of d AND half range to get 2% ✓	expect to see 80/80.2 and/or 1.5 Condone 1.9% to 2 sf	2	AO2 × 2

Question	Answers	Additional comments/Guidelines	Mark	AO
10.2	Best-fit line ✓		1	AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
10.3	Uses $E_p = F \times d$ to show $x^2 = \frac{2\mu mgd}{k}$ ✓		1	AO3

Question	Answers	Additional comments/Guidelines	Mark	AO
10.4	Use of large triangle to find gradient ✓ Correct use of formula ✓ $\mu = 0.29$ to 0.31 ✓	the minimum length of horizontal side is 7.0×10^{-2} Penalise POT error in mp3 only	3	AO3 × 2 AO2 × 1
Total			7	

Question	Key	Answer	AO

15	C		
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Question	Answers	Additional comments/Guidelines	Mark	AO
11.1	Uses $T \cos 35 = mg$ ✓ Use of $E = \frac{TL}{A\Delta L}$ or use of $E = \frac{\sigma}{\epsilon}$, $\sigma = \frac{F}{A}$ and $\epsilon = \frac{\Delta L}{L}$ ✓ Uses $A = \frac{\pi d^2}{4}$ or $A = \pi r^2$ ✓ 14(.2) (mm) ✓	Expect to see 916 or 920 (N)	4	$2 \times \text{AO2}$ $2 \times \text{AO3}$

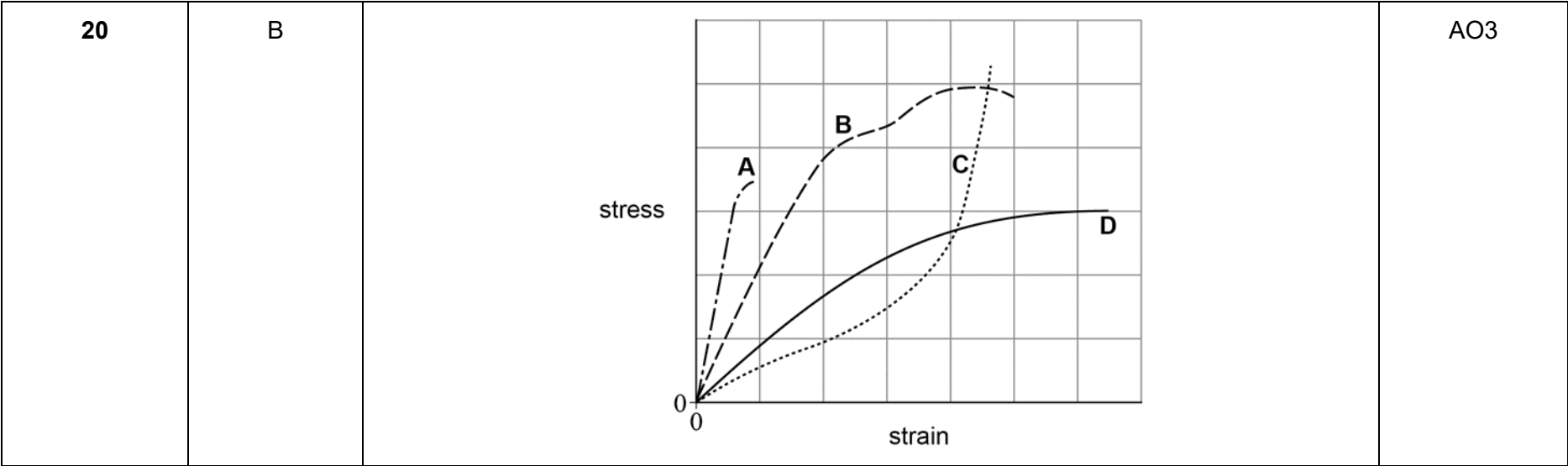
Question	Answers	Additional comments/Guidelines	Mark	AO
11.2	Idea that angle between the building and the rope decreases ✓ Idea that T is inversely proportional to $\cos\theta$ so as $\cos\theta$ increases T will decrease ✓ Idea that N is proportional to $\sin\theta$ and to T so as T and $\sin\theta$ decrease N will decrease ✓	MP1 can be expressed in either part of the answer Allow N is proportional to $\tan\theta$ or $1/\tan(90-\theta)$ Allow use of Pythagoras with a triangle of forces and a constant weight	3	$1 \times \text{AO1}$ $1 \times \text{AO2}$ $1 \times \text{AO3}$

Question	Answers	Additional comments/Guidelines	Mark	AO
11.3	Uses $F = k\Delta L$ to show that k is inversely related to ΔL (for a constant force) so extension is greater ✓ Idea that the change in momentum of the climber happens over a greater time so less force exerted (on the climber) ✓	Allow use of $\Delta GPE = 1/2 k\Delta L^2$ show that k is inversely related to ΔL (for an approximately constant GPE) so extension is greater Allow and work is done over a larger distance so less force exerted (on the climber) Allow an acceleration is lower so less force exerted (on the climber)	2	AO2
Total			9	

Question	Key	Answer	AO

12	A	$\frac{EA}{L}$	AO2
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19	C	0.18 m	AO2
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21	A	0.3%	AO2
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22	C	2.0×10^4	AO1
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Question	Answers	Additional comments/Guidelines	Mark	AO
01.1	The material is permanently deformed / does not return to its original length (when force is removed) ✓		1	AO1

Question	Answers	Additional comments/Guidelines	Mark	AO
01.2	Idea that the material undergoes little or no plastic deformation before failure ✓		1	AO1

Total			2	
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Question	Answers	Additional comments/Guidelines	Mark	AO
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Question	Answers	Additional comments/Guidelines	Mark	AO
03.1	kg s^{-2} ✓		1	AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
03.2	Attempts to calculate area under graph ✓ 30 to 33.5 (J) ✓ 31 to 32.5 (J) ✓✓		3	1 × AO1 2 × AO3

Question	Answers	Additional comments/Guidelines	Mark	AO
03.3	The stiffness of S increases as compression increases ✓ because the gradient of the line increases ✓	Condone spring constant for stiffness	2	AO3

Total			6	
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Question	Answers	Additional comments/Guidelines	Mark	AO
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Question	Answers	Additional comments/Guidelines	Mark	AO
08.1	Uses $\rho = \frac{m}{V}$ to calculate volume ✓ Uses $V = \pi r^2 h$ to calculate radius ✓ Doubles to give 76 mm ✓		3	1 × AO1 2 × AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
08.2	Uses equation of motion to determine a ✓ $mg \sin \theta$ seen ✓ Uses $F = ma$ ✓ $\theta = 33^\circ$ ✓ OR Uses $E_k = \frac{1}{2}mv^2$ ✓ Uses $W = \text{friction} \times s$ ✓ Uses $E_p = E_k + W$ ✓ Uses trig to determine $\theta = 33^\circ$ ✓	Expect to see $a = 2.9(0) \text{ (m s}^{-2}\text{)}$ Expect to see $\theta = \sin^{-1} \left(\frac{ma + R}{mg} \right)$	4	AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
08.3	Uses equation of motion ✓ to get $t = 4.6(4)$ (s) ✓		2	1 × AO1 1 × AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
08.4	(repeated) to minimise the effect of random errors ✓ (done in different locations) to take account of any lack of uniformity ✓	Allow 'the ground may not be uniform / horizontal'	2	AO4

Total			11	
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Question	Answers	Additional comments/Guidelines	Mark	AO
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Question	Answers	Additional comments/Guidelines	Mark	AO
10.1	Max two from: ✓✓ • calculates area in m^2 • use of stress = $\frac{\text{force}}{\text{area}}$ • uses Young modulus = $\frac{\text{stress}}{\text{strain}}$ 2.96 GPa ✓	$2.1 \times 10^{-6} \text{ m}^2$ correct power of ten must be seen	3	$2 \times \text{AO1}$ $1 \times \text{AO2}$

Question	Answers	Additional comments/Guidelines	Mark	AO
10.2	Uses trig to determine $\theta = 65^\circ$ ✓ Uses $F = 2T\cos 65$ to determine F ✓	expect to see 790 (N)	2	$1 \times \text{AO2}$ $1 \times \text{AO3}$

Question	Answers	Additional comments/Guidelines	Mark	AO
10.3	Any three from: ✓✓✓ • because $2T\cos\theta = ma$, acceleration decreases from an initial maximum • force decreases as extension/strain decreases • rate of decrease in extension/strain/force decreases • acceleration due to the tension is zero when the length is 1.48 m because the horizontal component of the tension is zero		3	$1 \times \text{AO2}$ $2 \times \text{AO3}$
Total			8	

Question	Key	Answer	AO
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Question	Answers	Additional comments/Guidelines	Mark	AO
09.1	Any two from: ✓✓ • correct use of $230 \sin 35$ or $2.4 \sin 35$ • attempts to apply principle of moments with correct distances • calculates mass of the bar using their weight 27 (kg) ✓	or $\cos 55$	3	AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
09.2	Calculation of cross-sectional area ($9.50331777 \times 10^{-5} \text{ m}^2$) OR Use of $E = \frac{FL}{A\Delta L}$ ✓ $6.1 \times 10^{-5} \text{ (m)}$ ✓	Allow a POT error in MP1	2	AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
09.3	$660 \text{ to } 680 \text{ (N)}$ ✓		1	AO1

Question	Answers	Additional comments/Guidelines	Mark	AO
09.4	Attempt to calculate an area under the graph ✓ linear unloading curve, starting at (8,1000) drawn on graph meeting the x axis between 5.5 and 6.5 mm ✓ Work done = 5 to 6 (J) from correct working ✓	Expect to see the area between the loading and unloading curves Allow 1 sf	3	AO3
Total			9	

Question	Answers	Additional comments/Guidelines	Mark	AO
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Question	Answers	Additional comments/Guidelines	Mark	AO
10.1	Uncertainty in L or $L_0 = 2 \times 0.5 (= 1 \text{ mm}) \checkmark_1$ Idea that subtraction of data leads to addition of uncertainties $\checkmark_2$ OR Idea that there are 4 readings taken (for one spring) so $4 \times 0.5 \text{ mm} \checkmark_1 \checkmark_2$		2	AO1

Question	Answers	Additional comments/Guidelines	Mark	AO
10.2	Correct steepest and shallowest lines with both lines going through all error bars. $\checkmark$	Steepest through right side of the second from bottom error bar and left side of the second from top error bar Shallowest through left side of bottom and right side of top Ignore addition line of best fit drawn	1	AO3

Question	Answers	Additional comments/Guidelines	Mark	AO
10.3	Calculates at least one gradient using correct method and triangle with minimum change in ΔL of 4 cm ✓ Attempts to find both gradients and calculates their mean ✓ $k_t = 0.81$ to 0.83 ✓	Expect to see: minimum gradient = 0.79 maximum gradient = 0.86 Do not allow “gradients” taken from points not on the line	3	AO3

Question	Answers	Additional comments/Guidelines	Mark	AO
10.4	uncertainty = $\frac{\text{max gradient} - \text{minimum gradient}}{2}$	Expect to see 0.035 Accept max-mean or mean-max	1	AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
10.5	ΔL decreases (for same F) OR F increases (for same ΔL) OR gradient is steeper (same error bar width) ✓ uncertainty increases / higher (for similar random error) ✓	MP2 dependent on MP1 Alternative: cannot say as uncertainty is based on scatter of points ✓✓	2	AO3

Total			9	
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Question	Answers	Additional comments/Guidelines	Mark	AO
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19	D	$2e$	AO1
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25	C	S has the largest spring constant.	AO1
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Question	Marking guidance	Mark	Comments
04	(proton) down 2/−2 ✓ (nucleon) down 4/−4 ✓	2	

Question	Marking guidance	Mark	Comments
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